



PUBLIC

SAP Digital Manufacturing in Regulated Environments

Quality and Compliance



Disclaimer:

Customers intending to use SAP Digital Manufacturing in a GxP context must independently assess the use of the [SAP Digital Manufacturing](#) product. SAP is a technology provider of business processes designed for general availability. SAP provides this document for informational purposes only. Any information outlined in this document may change at any time at SAP's discretion.

Contents

Abstract	4
1. Introduction to GxP and SAP Digital Manufacturing	4
1.1 Understanding GxP and GAMP 5 Framework	4
1.2 Overview of SAP Digital Manufacturing	5
1.2.1 DM Architectural Overview	5
1.2.2 SAP Digital Manufacturing Releases	6
1.3 SaaS Shared Responsibility Model	7
2. SAP Digital Manufacturing in GxP Regulated Environment	7
2.1 Certifications and Attestations	8
2.2 GxP SKU: Quality Requirement Schedule	8
2.2.1 Audit	9
2.2.2 Regulatory Requirements Traceability	9
2.2.3 Infrastructure Qualification	9
2.3 Testing Framework	9
2.4 Service Catalog	10
3. SAP Digital Manufacturing Quality Management System	10
3.1 Document Management	11
3.2 Cloud Service Qualification	11
3.3 Risk Management	11
3.4 Secure Software Development Operations Life Cycle (SSDOL)	12
3.5 SAP Digital Manufacturing IT Operations and Service	13
3.5.1 Incident Management	13
3.5.2 Problem Management	14
3.5.3 Operational Change Management	14
3.5.4 Monitoring Management	14
3.5.5 Backup and Restore	14
3.5.6 Provisioning and Deprovisioning	15
3.5.7 Hotfix	15
3.6 Security Management	15
3.7 Data Protection	16
3.8 Personnel Training	16
4. Next Steps	17
5. Conclusion	17
6. Glossary	17
7. History	20

Abstract

This whitepaper provides information on how SAP approaches GxP-related compliance for SAP Digital Manufacturing and provides customers guidance on using SAP Digital Manufacturing in a GxP context.

1. Introduction to GxP and SAP Digital Manufacturing

1.1 Understanding GxP and GAMP 5 Framework

In regulated industries such as pharmaceutical drugs manufacturing, medical devices, regulated Fast-moving Consumer Goods (FMCG), and chemicals, compliance with regulatory standards is critical to ensuring product quality and end-user safety. Collectively referred to as GxP, these “Good Practice” guidelines include Good Manufacturing Practices (GMP), Good Clinical Practices (GCP), Good Laboratory Practices (GLP), and others. Each framework sets standards for specific stages of the product lifecycle, from development to distribution.

Regulatory authorities such as the U.S. Food and Drug Administration (FDA), the European Medicines Agency (EMA), and others, provide guidelines that govern GxP practices. In the European Union and the UK, the General Data Protection Regulation (GDPR) and UK GDPR (enhanced by Data Protection Act 2018) mandates strict controls on personal data processing. In the United States, GxP compliance aligns with the Code of Federal Regulations (21 CFR), while in European Union, it aligns with EU Annex 11. These frameworks and guidelines underscore the importance of data protection mechanisms in regulated environments, where SAP provides support with comprehensive [technical and organizational controls](#).

Globally operated regulated industry organizations must harmonize with international guidelines, such as those developed by the International Council for Harmonization (ICH), World Health Organization (WHO), and International Organization for Standardization (ISO). Since there is no GxP certifications for cloud providers, GxP-regulated customers are responsible for verifying compliance across the full lifecycle of their GxP systems in the cloud. Customers operating in GxP context are ultimately accountable to regulatory bodies for demonstrating compliance with respective regulations.

The International Society for Pharmaceutical Engineering (ISPE) has collaborated with regulatory authorities and industry experts to develop the Good Automated Manufacturing Practices (GAMP 5) framework. GAMP 5: A Risk-Based Approach to Compliant GxP Computerized Systems provides guidelines for managing software lifecycle based on good practices. The GAMP guidelines under GAMP 5 and GAMP Good Practice Guide: IT Infrastructure Control and Compliance cover cloud-based Infrastructure and software delivery models, including SaaS.

GAMP 5 guidance encourages regulated companies to maximize supplier involvement throughout the system life cycle, leveraging the supplier's knowledge, experience, and documentation where possible. This requires a satisfactory supplier assessment to establish trust in SAP's processes and controls. A thorough supplier assessment is essential to verify that SAP Digital Manufacturing, develops and manages according to a Quality Management System that follows documented procedures and standards.

This whitepaper provides customers an overview of SAP practices and capabilities supporting usage of SAP Digital Manufacturing in a GxP context. This information can be used to gain deeper

understanding as a part of supplier assessment. SAP summarizes specific GxP-relevant information for SAP Digital Manufacturing, including:

- The technical and procedural measures and controls
- Its institutional knowledge to develop and operate SAP Digital Manufacturing in a controlled state
- Methods to ensure the confidentiality, integrity, and availability of customers' data

This Whitepaper consolidates and highlights GxP-relevant aspects, like SAP Digital Manufacturing's Quality Management System (QMS), Secure Software Development Lifecycle (Secure SDL), and IT Operations and Services.

1.2 Overview of SAP Digital Manufacturing

SAP Digital Manufacturing is a multi-tenant Software as a Service (SaaS) solution, delivered over internet. SAP manages both the infrastructure and the application, controlling more of the technology stack and delivery cycle than in traditional on-premises models. This paradigm shifts on technology and delivery ownership required customers to adapt their GxP compliance strategies.

As a cloud-based Manufacturing Execution System (MES), SAP Digital Manufacturing enables enterprises to manage and optimize manufacturing operations in real time. The solution is designed to drive shop floor execution and enhance production efficiency. Delivered as a Software-as-a-Service (SaaS) application, SAP DM combines scalability, flexibility, and continuous innovation to meet the demands of modern, connected manufacturing. With a SaaS delivery model customers trade capital expense for variable expense, while receiving the latest innovations.

In the following sections, we will first explore SAP Digital Manufacturing's architectural design and its agile release cycle, then we will have a deeper dive into detailed explanations on how SAP DM supports compliance in a GxP regulated environment.

1.2.1 DM Architectural Overview

SAP Digital Manufacturing runs on SAP's Business Technology Platform (BTP), a Platform as a Service (PaaS), computing environment hosted on hyperscaler infrastructures (IaaS). See the illustration below for more information.

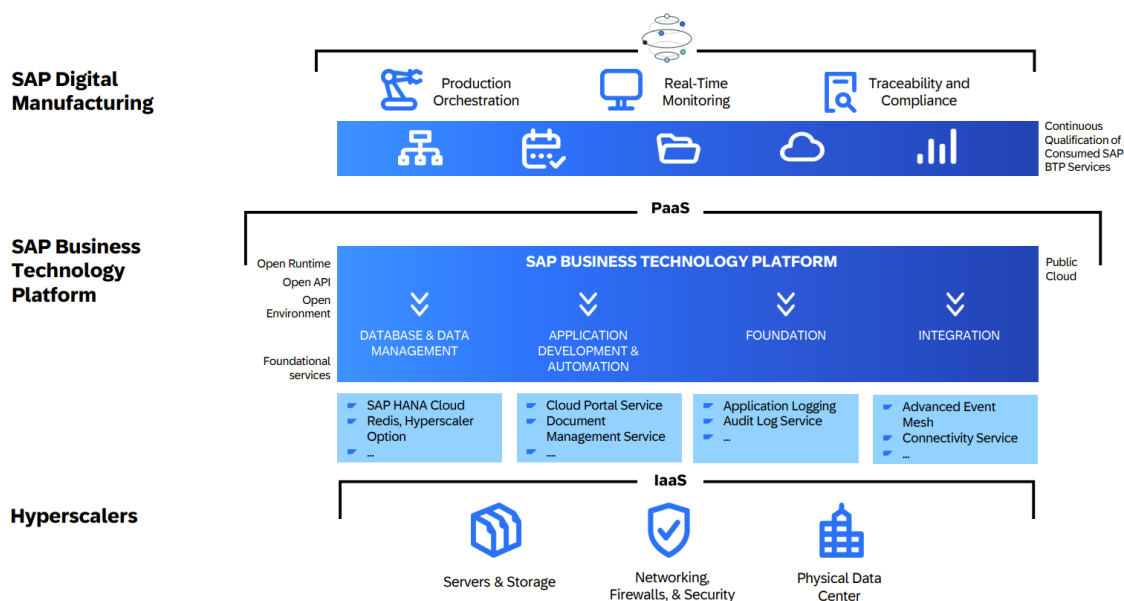


Figure 1: SAP Digital Manufacturing Architecture

The underlying infrastructure is managed by the underlying platform (SAP BTP) as a part of their infrastructure management practices detailed in [SAP Business Technology Platform Services in GxP Environments](#). SAP BTP is responsible for the SAP Cloud Infrastructure and the underlying operating systems. This responsibility includes protecting customer systems and data from threats originating from other cloud tenants or networks, and ensuring the security of the infrastructure. Since only non-SAP data centers provided by hyperscalers are used, back-to-back agreements are in place by SAP BTP to guarantee the same level of security and compliance as described in details in the [SAP BTP whitepaper for GxP environments](#).

1.2.2 SAP Digital Manufacturing Releases

SAP Digital Manufacturing follows quarterly release cycles. Updates are first deployed to the Quality environment, enabling customers to test and validate changes before their promotion to the Production environment. SAP Digital Manufacturing provides user assistance documentations, which includes standard Release Notes, [What's New Viewer](#) and a [Service Catalog](#). The Service Catalog includes unique identifiers for all changes to help customers to trace the changes associated with every release. Release schedules, including publication dates for What's New Information, are available on [SAP Help Portal](#). The promotion of code to the production environment follows release decision criteria. For additional details, refer to section 3.4 on the Secure Software Delivery Lifecycle.

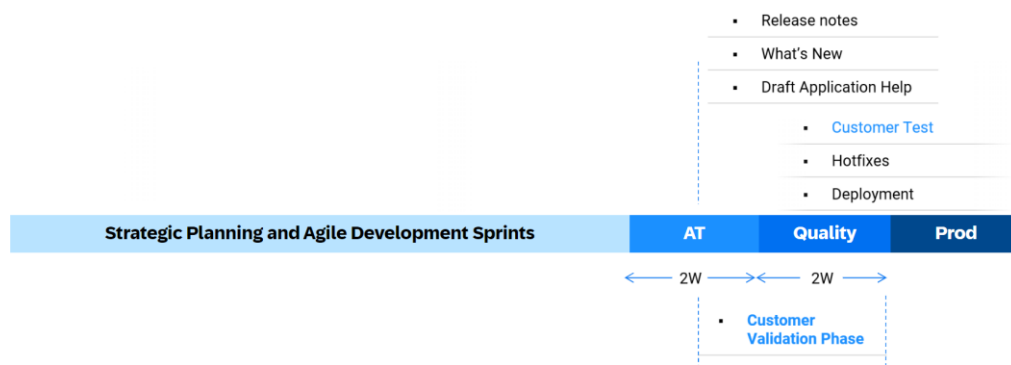


Figure 2: SAP Digital Manufacturing Release Cycle

1.3 SaaS Shared Responsibility Model

A key concern for SAP Digital Manufacturing customers operating in GxP environment is how to qualify the system and demonstrate control, given the shared responsibilities between the customer and SAP. The shared responsibility, shown in the model below, is a significant shift from the on-premises delivery model. In this model for SaaS delivery, SAP is responsible for software development, operations, and infrastructure management (through IaaS providers), alongside platform maintenance and implementation of security controls. Meanwhile customer retains responsibilities including validation of system's fitness for the intended use, system qualification documentation, and demonstrating compliance of GxP practices to the relevant regulatory bodies.

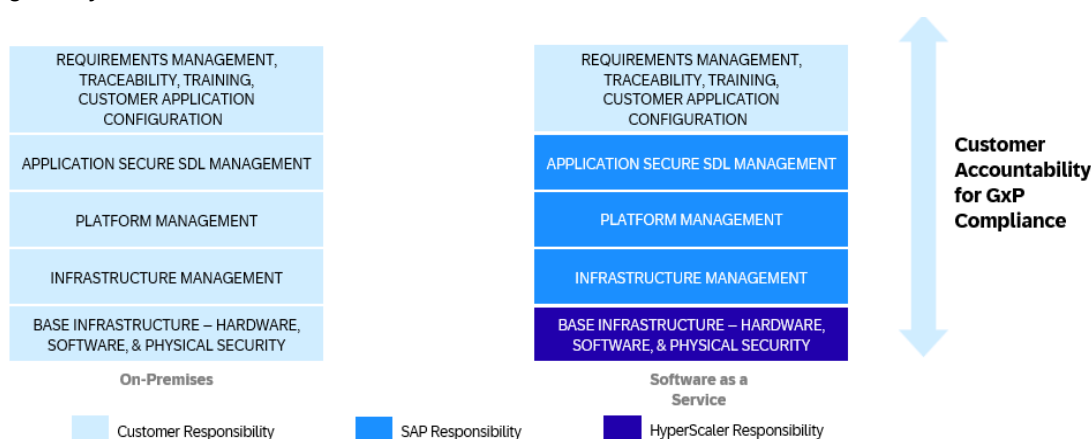


Figure 3: Responsibilities in On-Premises vs SaaS

2. SAP Digital Manufacturing in GxP Regulated Environment

SAP Digital Manufacturing provides regulated customers a building block approach to establish their GxP strategy that leverages SAP's artifacts and offerings. Through this approach customers can establish a qualification strategy and operating model for SAP Digital Manufacturing while being aligned to their internal quality management systems. Our approach

provides capabilities in the following areas: Auditability, Traceability, Infrastructure Qualification and Testing Framework.



Figure 4: GxP Offerings from SAP Digital Manufacturing

2.1 Certifications and Attestations

SAP supports specific certifications and attestations to achieve specific assurances. The development and support of SAP Digital Manufacturing is in scope of the ISO 9001 Certification for development. The underlying BTP holds SOC1, SOC2, ISO 27001 and ISO 9001 certifications and attestations. Regulated industry customers can find the latest attestations and certifications on the [Trust Center, Compliance section](#).

Note: SAP Digital Manufacturing now supports integration of a generative AI copilot – Joule. Joule is undergoing SOC audits and not yet covered by above mentioned certification and attestation.

2.2 GxP SKU: Quality Requirement Schedule

SAP offers a dedicated GxP SKU, known as the SAP Quality Requirement Schedule, to support customers in regulated environment. The SAP Quality Requirement Schedule focuses on providing customers with:

1. The Right to Audit: Customers gain the right to audit SAP Digital Manufacturing's Quality Management System (QMS) and related Change Management documentation under specific terms and conditions.
2. Regulatory Requirement Traceability Matrix (RRTM): A comprehensive matrix that helps map regulatory requirements to SAP's operational controls, facilitating customer's efforts to align their use of SAP Digital Manufacturing with applicable GxP standards.
3. Installation Qualification: Documentation supporting the installation of the customer tenant and the continuous qualification of underlying platform (BTP).

From June 2024 onwards, the SAP Quality Requirement Schedule was made available to provide customers in regulated industries with a mechanism to leverage SAP Digital Manufacturing's existing processes and documentation.

Although the responsibility for compliance with GxP regulations lies with the regulated company, the supplier may have considerable involvement in the process.

Regulated companies may be able to leverage supplier knowledge and documentation, subject to suitability following formal assessment. This may involve an audit, depending on risk, complexity, and novelty.

– GAMP 5

It is important to note that while SAP Digital Manufacturing provides extensive resources, we do not offer legal advice. Validation is ultimately the responsibility of the customer. Our commitment to GxP compliance is ongoing. For more details on the SAP Quality Requirement Schedule and its scope, please see sections below or refer to [SAP Help Portal](#).

2.2.1 Audit

The SAP Quality Requirement Schedule for Digital Manufacturing subscription includes the right to audit the Quality Management System and Change Management documentation related to SAP Digital Manufacturing. Due to the confidential nature of the documentation shared in the audit, the audits must be conducted on-site. With these on-site audits, customers can verify and confirm if the control activities related to SAP Quality Management System (QMS) meet their intended purpose.

2.2.2 Regulatory Requirements Traceability

SAP Digital Manufacturing provides customers subscribing to SAP Quality Requirement Schedule with a Regulatory Requirements Traceability Matrix (RRTM). This matrix maps 21 CFR Part 11 and EU Annex 11 regulatory requirements to features within SAP Digital Manufacturing and SAP BTP. Customers often need a similar document to record the extent to which regulatory requirements have been fulfilled and to demonstrate compliance. This RRTM serves as a preliminary tool for customers to establish traceability between SAP Digital Manufacturing and the applicable regulatory requirements. The RRTM document is updated with release of new version of SAP Digital Manufacturing.

2.2.3 Infrastructure Qualification

As part of the infrastructure/platform qualification, SAP Digital Manufacturing provides customers subscribing to SAP Quality Requirement Schedule with an Infrastructure Qualification Precursor Document. This document can be used as a verification tool, containing all necessary evidence in the form of screenshots from our internal dashboard to confirm the successful creation of customer tenants, in addition to the email confirmation customers receive after the setup of the instance.

Furthermore, as Digital Manufacturing operates on SAP Business Technology Platform (BTP), the qualification of BTP services is essential for regulatory compliance. SAP continuously qualifies the underlying BTP services utilized by SAP Digital Manufacturing. Customers are granted on-site audit access to review these Continuous Service Qualification (CSQ) reports.

2.3 Testing Framework

GAMP 5 advocates a computerized system validation framework to achieve and maintain GxP compliance throughout the computerized system life cycle. The framework emphasizes the usage of automation to achieve greater control, higher quality, and lower risks throughout the system lifecycle. For SAP Digital Manufacturing, automated testing/validation is key to demonstrating GxP compliance, due to its SaaS delivery model. With quarterly release every three months and a small validation window (2-3 weeks) traditional manual validation approaches may prove insufficient.

By implementing automated validation process, customers can significantly accelerate their testing cycles for new SAP Digital Manufacturing releases and demonstrate compliance. SAP offers Enterprise Support customers with free access to Tricentis [Test Automation](#). Customers can use this testing automation tooling to establish automated computerized system validation for SAP Digital Manufacturing. Furthermore, SAP also provides test automation sample modules based on Tricentis Test Automation, so that customers can build their automated test cases quickly.

2.4 Service Catalog

SAP provides a [Service Catalog](#), and [What's New Viewer](#), updated with every release on the SAP Help Portal. The service catalog maintains an inventory of all Apps, Pods, Plugins, and APIs, with each element assigned a unique identifier for traceability. The What's New Viewer for SAP Digital Manufacturing, includes a description of all new, enhanced, or changed features together with reference to the service catalog IDs.

Customers can leverage the service catalog for their functional specification documentation. Both the service catalog and the what's new viewer can be used to know what is changed with every new release and establish traceability to assess the impact of those changes.

3. SAP Digital Manufacturing Quality Management System

The SAP Global Quality Policy establishes the overarching principles for the Quality Management System (QMS) utilized across SAP, including those supporting development of SAP Digital Manufacturing. While SAP Digital Manufacturing aligns with SAP's global QMS, it also operates its own incremental Standard Operating Procedures (SOPs), and Memos tailored to the quality related activities performed by SAP Digital Manufacturing team.

SAP Digital Manufacturing defines processes for customer requirement management, and product delivery, while establishing clear responsibilities between different roles. The processes oversee deliverables, documentation, and audit mechanisms, handles incidents, and drives continuous improvement through adoption of modern development practices. This structure fosters accountability and streamlined design-making across various levels of SAP Digital Manufacturing operations. SAP Digital Manufacturing adopts a set of documented Standard Operating Procedures (SOPs), and Memos which standardize and guide operational processes to ensure consistent adherence to quality principles.

Again, while SAP Digital Manufacturing's SOPs address its operational requirements, the solution is firmly integrated with SAP's overarching global QMS, which provides broader governance and enterprise-wide standards.

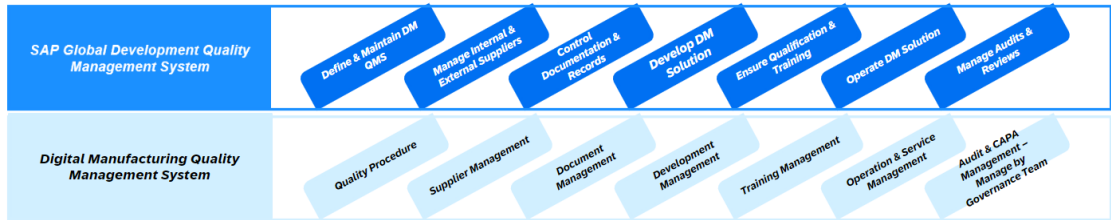


Figure 5: SAP Global Development QMS and SAP Digital Manufacturing SOPs

3.1 Document Management

SAP Digital Manufacturing maintains its SOPs, Memos and other relevant documents in a controlled electronic Document Management System (eDMS), a secure platform equipped with automated version control, user role management, and e-signature functionality. Access to the eDMS is restricted to authorized personnel based on defined roles and responsibilities, safeguarding the integrity and confidentiality of the stored documents.

To ensure accuracy and relevance, all SOPs, Memos and relevant documents undergo a structured review and approval process following SAP Digital Manufacturing's documented procedure. Document owners and reviewers are responsible for maintaining the correctness, completeness, and alignment of SOP content with applicable policies and processes. The Document review and approval is carried through eDMS which also records the associated electronic signatures.

Approved versions of SOPs are securely stored in the eDMS, where they remain accessible to authorized users for operational needs. The eDMS also supports document version control, ensuring that only the latest approved versions of SOPs are available for use while archiving previous versions for traceability.

3.2 Cloud Service Qualification

SAP Digital Manufacturing is a multi-tenant SaaS solution built on the platform of SAP BTP. A risk-based approach of continuous "Cloud Service Qualification" (CSQ) is used to support the Infrastructure Qualification for SaaS based SAP Digital Manufacturing. This approach mitigates the risk of unintended changes at the underlying platform level based on continuous verification that the intended use of the leveraged BTP services is still met during update cycles.

The Cloud Service Qualification follows a defined process that begins with assessment of the application, functional services, and service risks. These assessments then inform the creation of a service qualification plan, which concludes with the generation of qualification reports. The entire documentation can be reviewed by customers subscribing to SAP Quality Requirement Schedule during an audit.

3.3 Risk Management

SAP's risk management policy provides a structured approach to identifying, prioritizing, and directing risk management activities for SAP Digital Manufacturing. Risk management compliance in SAP implies an effective early-warning system and a holistic risk management program including operational, financial, compliance, and strategic risk management.

SAP maintains an effective and efficient risk management system to create risk awareness within the company, to enable the business in informed decision-making, to successfully achieve the company's goals, and to ensure customer satisfaction as well as legal and regulatory compliance.

The risk management process is based on the following steps:

1. Risk Planning
2. Risk Identification
3. Risk Analysis
4. Risk Response
5. Risk Monitoring
6. Risk Validation

3.4 Secure Software Development Lifecycle (Secure SDL)

The secure software development lifecycle of SAP Digital Manufacturing is aligned with SAP's broader QMS and product standards. These standards guide the development process to meet essential quality benchmarks, including functional correctness, security, performance, and user experience.

SAP Digital Manufacturing follows an agile development methodology to iteratively design, develop, test, and deliver new features. Requirements (both functional and non-functional) undergo a refinement process and are broken down into backlog items in SAP's project management tooling in the form of Epics, User Stories, and Tasks.

The development is done in sprints, with code managed in a central repository. Every change (Pull request) to the development code is reviewed and tested. The development process incorporates testing at multiple levels, including unit, integration, and regression tests, to ensure code quality and functionality. The development process follows continuous integration and continuous deployment principal. Quality gates are integrated into the development process to assess code readiness when promoting code to higher environments.

All activities from requirements to development to final release are maintained and tracked in SAP's project management tooling. This allows traceability of artifacts and ensures transparency and accountability.

After the closure of development phase, SAP conducts acceptance testing to verify release readiness. This process evaluates system integration, performance, and operational readiness. Once acceptance testing phase is successfully completed the code is deployed to sap internal pre-production and customer test environment. Upon successful internal testing, verification and customer testing the code is deployed to Production environment. For each release, SAP also provides supporting documentation, such as feature descriptions and release notes, to help customers stay informed.

SAP's software delivery lifecycle is structured to enable continuous improvement. Defects identified during testing or after deployment are analyzed for root causes, and corrective measures are implemented to prevent recurrence. These learnings contribute to refining the overall development and testing strategy, ensuring SAP Digital Manufacturing evolves to meet customer expectations and maintain standards.

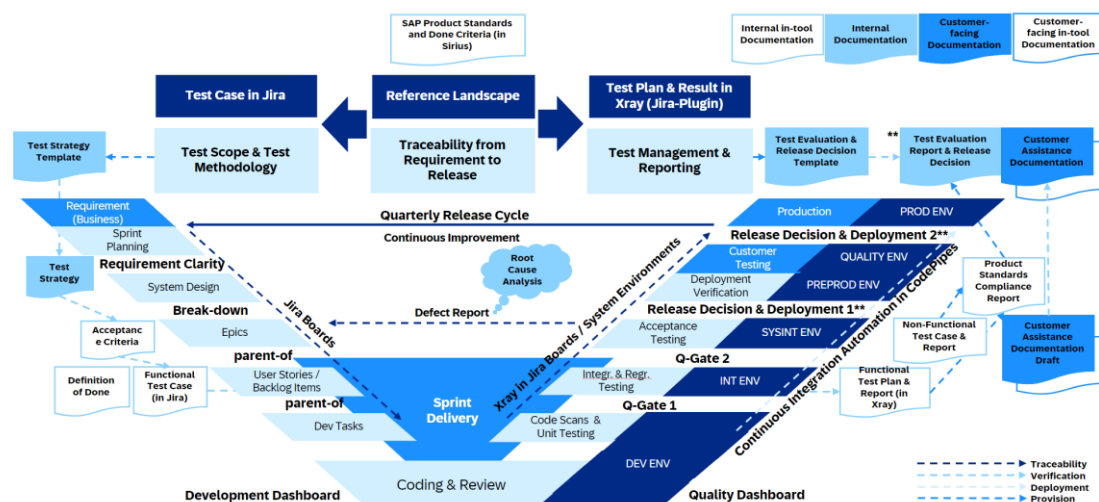


Figure 6: SAP Digital Manufacturing Product Qualification Model

3.5 SAP Digital Manufacturing IT Operations and Service

SAP Digital Manufacturing has comprehensive procedures for maintaining control of its system throughout its operational life. The standard operating procedures (SOP) for SAP Manufacturing IT Operations and Services are:

- Operational Change Management
- Monitoring Management
- Backup and Restore
- Provisioning and De-Provisioning
- Hotfix Release Process
- Incident Management (SAP Global)
- Problem Management (SAP Global)

The above SOP for SAP IT Operations for the SAP Manufacturing that applies to SAP Manufacturing Edge on-premises component, as an add-on or a supplement, are only:

- Incident Management*
- Problem Management*

*The limited SOP is because the Life Science Customer runs and operates SAP Digital Manufacturing Edge on its shop floor; thus, it is responsible for the provisioning, installation, and qualification required for the on-premises IT infrastructure. The Life Science Customer is also responsible for the secure operation, including specific backup and restoration responsibilities and change management of SAP Digital Manufacturing Edge. See Figure 4 above for reference.

3.5.1 Incident Management

An operation incident refers to any unplanned event that prevents (or may prevent) or delays users, the system, an operation, or service from proceeding with an assigned task. The Incident management procedure aims to categorize Incidents to direct them to the most appropriate resource or complementary process to achieve a timely resolution. As a general guiding principle, SAP fixes priority Incidents opened against the production environment within defined timelines as specified in service level agreements (SLA). If necessary, such resolutions may lead to a Hotfix, see section 3.5.7.

SAP Digital Manufacturing follows SAP's global standards on Incident Management. The SAP Digital Manufacturing operations support team handles customer support, including customer/internal ticket processing, service requests, addressing service availability or degradation issues.

For Incident management, SAP Digital Manufacturing follows the following process steps:

1. Log an Incident
2. Categorize an Incident
3. Investigate and Diagnose
4. Resolve and Recover
5. Validate and Close

The Support Policy for SAP Cloud Services outlines (i) information on how to report/log an Incident and (ii) customer response levels, including priority, definitions, initial response times, ongoing communication objectives, and resolution target.

3.5.2 Problem Management

SAP Digital Manufacturing's problem management process addresses the lifecycle of issue tickets created either automatically or manually created. Its objective is to determine root causes, prevent recurring future incidents, reduce recurring customer system outages, and minimize incidents' business impact.

SAP Digital Manufacturing follows the Problem Management procedure with the following steps:

1. Log and Categorize a Problem
2. Diagnose and Define a Problem
3. Resolve and Realize a Problem
4. Validate and Close a Problem

3.5.3 Operational Change Management

SAP Digital Manufacturing follows an operational change management process with the following process steps:

1. SAP maintains an inventory of application lifecycle management tools and services, listing the relevant software components with all necessary attributes.
2. SAP monitors upcoming component versions. There is a software component version upgrade plan for each use to ensure all software components operate under the latest version.
3. SAP implements new component versions with quarterly releases. The implementation of component upgrades follows the SDOL.

In the scope of Operational Change Management are (i) upgrades (regular product increments) and (ii) corrections to the infrastructure and platform components, systems, and services, including upgrades of the platform, infrastructure components, database, runtime environment, and operating system.

Changes to the SAP Digital Manufacturing solution are not in the scope of this process. The development of quarterly SAP Digital Manufacturing releases follows the SAP Digital Manufacturing release process in the scope of SDOL.

3.5.4 Monitoring Management

The monitoring management process focuses on the monitoring and reporting of the availability of the SAP Digital Manufacturing application, as well as handling of availability incidents. The objective of monitoring availability is to enable the delivery of consistent and timely service to SAP Digital Manufacturing customers.

The Service Level Agreement for SAP Cloud Services outlines the system availability SLA and the calculation/definition of the system availability percentage. SAP conducts Monitoring Availability for SAP Digital Manufacturing using the SAP BTP availability service, an SAP internal tool to monitor, analyze, alert, and report on the availability of services.

Key monitoring availability details, including incident times, availability status, and root-cause analyses, are reported in the SAP Trust Center under the Cloud Availability Center. This reporting is specific to data centers and publicly accessible.

3.5.5 Backup and Restore

The Backup and Restore process safeguards against data loss and support the restoration of digital assets when necessary. This process applies to the SAP Digital Manufacturing persistence platform services. SAP HANA Cloud (SAP HANA DB), a database-as-a-Service offering managed

by BTP, is the data storage for tenant-specific relational business data. BTP also manages backups, patch management, and other platform service needs.

3.5.6 Provisioning and Deprovisioning

Provisioning ensures the proper and secure set-up of a customer tenant, starting with onboarding once a customer purchases SAP Digital Manufacturing. Deprovisioning ensures the proper and secure deletion of customer data upon contract termination.

The SAP Help Portal describes the onboarding process and required steps for SAP Digital Manufacturing. The onboarding process starts after the customer purchases SAP Digital Manufacturing. During the process, the customer administrator receives emails from SAP with instructions for the next steps. If the customer does not already have a BTP global account or an SAP Cloud Identity Services - Identity Authentication tenant, the contact person is automatically assigned as the first administrator for the global account or tenant.

SAP triggers the customer system/tenant decommissioning upon receipt of the customer's request for termination or an event for the effective termination of the contract. SAP sends its customers a notification of the effective termination date upon this decommissioning. The customer must submit a service ticket to commence the data retrieval process during this notification period and has until the effective termination date to retrieve all data. If more time is needed, the customer may request an extension subject to a pro-rata fee for extended winddown activities. After the effective termination date, SAP continues completing the decommissioning process. SAP deactivates customer accounts and uses technical tools to delete customer data. The customer must retrieve all customer data before the effective termination date. SAP deletes all customer data remaining at the end of the decommissioning period, eliminating the possibility of restoring or retrieving such data.

3.5.7 Hotfix

Hotfixes are implemented to provide solutions for urgent customer issues or to address high-priority vulnerabilities. The process begins with SAP initiating the Hotfix through Incident Management, Problem Management, and Change Management workflows. A Hotfix is specifically required when resolving a customer or internal incident necessitates a code change.

Each Hotfix undergoes a Root Cause Analysis (RCA) and impact evaluation to ensure proper implementation. Following QA tests and internal approval, which includes a review of the RCA, impact analysis, and test results, the Hotfix is cleared for deployment. Once released, SAP publishes an accompanying Hotfix Release Note detailing the changes.

3.6 Security Management

SAP Digital Manufacturing operates under an Information Security Management System (ISMS) per ISO 27001.

SAP applies its Global Security Policy across all businesses, employees, external parties granted access to SAP Information, and information assets under its administration. This policy establishes company-wide requirements for the protection of SAP's personnel, operations, and customer information.

SAP Global Security policy defines the high-level requirements for:

- Preserving the confidentiality, integrity, and availability of information and assets
- Protecting assets from threats, whether internal or external, deliberate or accidental, based on an assessment of risks to the organization.

SAP Security Policy is supported by various standards, procedures, and good practices. Security controls are described in the applicable SOC1/SOC2 reports, which are available upon request via the SAP Trust Center.

3.7 Data Protection

SAP Digital Manufacturing implements processes, tools, and procedures to protect records, ensuring their accurate and ready retrieval. These measures include:

- Data Segregation
- Data Encryption
- Data Backup and Restore

Data Segregation: SAP supports the multi-tenant architecture of SAP Digital Manufacturing, which supports security, confidentiality, privacy, integrity, and availability of data standards. SAP Digital Manufacturing was designed with the assumption that all tenants are potentially hostile to all other tenants. Security measures to prevent the actions of one tenant from affecting the security or service of another tenant or accessing the content of another tenant are in place.

The two primary goals of maintaining tenant isolation are:

- Prevent leakage of, or unauthorized access to, customer data or content across tenants
- Prevent actions of one tenant adversely affecting the service for another tenant

SAP Digital Manufacturing implemented multiple forms of protection to prevent customers from compromising SAP Digital Manufacturing services or applications or gaining unauthorized access to the information of other tenants including logical isolation of customer content within each tenant.

Data Encryption: SAP Digital Manufacturing implemented encryption methods, protocols, and algorithms to help provide a secure path for data in transit through the SAP Digital Manufacturing infrastructure, and to help protect the confidentiality of data that is stored within SAP Digital Manufacturing.

Data Backup and Restore for the SAP Digital Manufacturing persistence services is described in section 3.5.5.

3.8 Personnel Training

GxP regulations mandate adequate training and education for personnel managing computerized systems in GxP-regulated contexts.

SAP Digital Manufacturing implemented a training program to ensure that employees for SAP Digital Manufacturing are adequately trained on applicable GxP processes and are qualified to perform their duties. New employees receive onboarding- and predetermined training requirements. SAP communicates applicable GxP policies and standards through mandatory trainings aimed at employees and external parties responsible for supporting SAP Digital Manufacturing.

SAP uses an internal Learning Management System (LMS) to manage critical course content and training traceability. This LMS includes a dashboard and reporting capabilities for managers to see overall training completion.

4. Next Steps

Achieving GxP compliance when adopting cloud-based solution is a journey. SAP has helped many customers along this journey.

Regulated companies require a high level of confidence that computerized systems will meet their technical, commercial, and regulatory requirements. They also wish to leverage the knowledge, experience, and documentation of the supplier.
– GAMP 5

To gain trust on SAP is critical for customers as they inherit certain controls related to infrastructure, application, and security. The customers can take the following approach towards performing a supplier assessment as a part of establishing the trust:

Basic Assessment: Customers can perform a basic supplier assessment based on publicly available information such as certification, market reputation, whitepapers etc.

Postal Audit using a questionnaire: Additionally, customers can ask clarifying questions in the form of supplier questionnaire, for supplementing the gathered information. Customers can reach out to their account team with the questionnaire.

On-site audit: Customers subscribing to the **SAP Quality Requirement Schedule**, detailed in Section 2.2, can request an on-site audit. The conditions of audit are described in more details in the SAP Quality Requirement Schedule contractual agreement. A supplier assessment often includes an in-detail review of SAP's QMS and operational practices, including certifications, audit reports, and whitepapers. Based on the shared responsibility model (see Section 1.3), SAP shares its security and control environment by doing the following:

- Obtaining and maintaining industry certifications.
- Publishing information about the SAP and Digital Manufacturing security and control practices in whitepapers.
- Provide certificates, reports, and other documentation directly to SAP customers, as needed.

5. Conclusion

SAP Digital Manufacturing is committed to providing a GxP capable solution that enables customers to meet their compliance requirements efficiently. Our extensive documentation and pragmatic approach help ease customer's validation and testing efforts while maintaining high quality standards. While this overview demonstrates our commitment to GxP capabilities, customers should consult their legal and compliance teams for specific regulatory requirements. Through continuous improvement and risk-based approaches, we partner with customers to achieve both compliance objectives and operational excellence in manufacturing.

6. Glossary

Abbreviation /Term	Definition
BTP	Business Technology Platform
CSNS	Cloud System Notification System

Abbreviation /Term	Definition
CSV	Computer System Validation
DBaaS	Database as a Service
DMS	Document Management System
EU	European Union
GAMP®	ISPE: A Risk-based Approach to Compliant GxP Computerized Systems
GDPR	General Data Protection Regulation
GxP	Good Practices for x = C (Clinical), D (Distribution), Doc (Documentation), L (Laboratory), M (Manufacturing), etc.
IaaS	Infrastructure as a Service
ISMS	Information Security Management System
ISO/IEC 27001	International Standard on Information Security Management
ISPE	International Society for Pharmaceutical Engineering
ITSM	IT Service Management
LMS	Learning Management System
MES	Manufacturing Execution System
PaaS	Platform as a Service
QA	Quality Assurance
QMS	Quality Management System
RDM	Release Decision Meeting
RCA	Root Cause Analysis
RPO	Recovery Point Objective
RTC	Release to Customer
SaaS	Software as a Service
SAP HANA DB	SAP HANA Database

Abbreviation /Term	Definition
SDLC	Software Development Life Cycle
SLA	Service Level Agreement
SOC	Service Organization Control
SOP	Standard Operating Procedure
TLS	Transport Layer Security

7. History

Version	Date	Change Description
1.0	26.07.2023	First Version
2.0	08.07.2024	Second Version: Adding new section [GxP SKU: Quality Requirement Schedule]
3.0	15.08.2024	Third Version: Adding Joule Integration Information
4.0	20.01.2025	Fourth Version: Revising and updating document to reflect comprehensive changes and enhancements